# Lost Link

HW7 | Benchmark Your Agentic System

MIT MAS.665 / November 4, 2026
**Team:** Kemeng Zhang, Yicheng Liu

## Claim and pre-set success check

For agents searching campus lost-and-found reports, vocabulary normalization should improve target retrieval over exact-category keyword search. We test both retrieval coverage and whether the service abstains when the requested item is absent.

**Primary metric:** correct retrieval-or-abstention per unique query. If a labeled target exists, it must appear in the top 3; if no target exists, the candidate list must be empty. These checks were saved before execution. Answer labels stay in the evaluator and are never supplied to either ranker. A hit is a suggestion for review, not proof of ownership.

## Fair baseline and experimental setup

The baseline is the repository's simpler exact-category filter plus raw keyword overlap. Original Lost Link adds category/color/location aliases, weighted scores and neutral credit for unknown fields. The revised arm changes only candidate eligibility. All three use the same 21 found reports, public inputs, API/auth/audit code, SQLite database, k=3 and machine; only the ranker is swapped.

Two deterministic authenticated clients stand in for agents: a holder registers found reports and an owner registers lost reports and searches through FastAPI TestClient. These are real ASGI requests and SQLite writes, with no LLM. The experiment isolates the service's retrieval benefit to agents. It does not evaluate autonomous planning or the return workflow.

## Results: unique cases, not repeated timing rows

| Check | Baseline | Original | Revised |
|---|---|---|---|
| Legacy positives: hit@3 | 3/10 | 10/10 | 10/10 |
| Tuning: primary success | 4/4 | 1/4 | 4/4 |
| Fresh positives: hit@3 | 4/6 | 6/6 | 5/6 |
| Fresh absent items: false suggestions | 0/6 | 5/6 | 0/6 |
| Fresh: primary success | 10/12 (83.3%) | 7/12 (58.3%) | 11/12 (91.7%) |

**Interpretation:** normalization improves coverage on known synonyms, but the original policy over-suggests on absent, unfamiliar types. The revision improves the fresh composite result by four cases over the original and one case over baseline, while losing one real target. This small sample does not establish statistical superiority.

## Team contributions

**Kemeng Zhang**: defined the claim, primary metric and pre-set success checks; designed the baseline and three-arm comparison; wrote the tuning and fresh test cases; diagnosed the original ranker’s false suggestions on absent items and designed the category-eligibility change; analyzed results and limitations; wrote the report.

**Yicheng Liu**: implemented the benchmark harness and verifier; ran the discovery and retest experiments; packaged the evidence bundle and reproducibility materials.

# A failure, one change, a fresh check

## Discovery and change

Four tuning cases vary the amount of type evidence: an absent ring, an absent scarf, an absent bookmark with all metadata missing, and a real bottle described as a flask with category left blank. The original returned wrong candidates for all three absent items; baseline abstained. On T03, category/color/location/date unknown-credit alone summed to the 0.45 threshold, producing F01/F02/F03 with no shared words. T01's unrelated bottle scored 0.713.

**One change:** require a non-null normalized query category and the same normalized category in a found report before applying the unchanged scoring function. No aliases, weights or thresholds were tuned. Type words in the description still count. Under identical inputs, tuning success improved from 1/4 to 4/4. Source and fixture hashes were frozen before fresh execution.

## All 12 fresh cases (P = passes the pre-set check; F = fails)

| Case / condition | Target | Baseline | Original | Revised |
|---|---|---|---|---|
| H01 Thermos / bottle synonym | F01 | F | P | P |
| H02 Phone / transparent cover | F02 | P | P | P |
| H03 Keys / red carabiner | F04 | P | P | P |
| H04 Brolly only in description | F06 | P | P | P |
| H05 Rucksack / backpack synonym | F09 | F | P | P |
| H06 Watch: unsupported true type | HW7-WATCH | P | P | F |
| H07 Ring absent | absent | P | F | P |
| H08 Scarf absent | absent | P | F | P |
| H09 Helmet absent | absent | P | F | P |
| H10 Violin absent | absent | P | F | P |
| H11 Calculator absent (known type) | absent | P | P | P |
| H12 Flash drive absent; metadata unknown | absent | P | F | P |

**The remaining failure is informative:** H06 has a real red watch in the database. Baseline and original retrieve it; the revised arm returns nothing because 'watch' is outside the category vocabulary. The change reduces false suggestions by accepting a coverage loss. Even the revised arm returns 0.417 wrong candidates per fresh query on average, so ownership still needs verification.

# Cost, evidence and limits

## Measured resources and run count

| Fresh search resources | Baseline | Original | Revised |
|---|---|---|---|
| API latency p50 / p95 (ms) | 1.131 / 1.341 | 13.417 / 16.589 | 4.524 / 9.399 |
| Mean response bytes | 372.8 | 1225.3 | 623.2 |
| Mean wrong candidates / query | 0.333 | 1.75 | 0.417 |

26 unique cases: 10 legacy regression queries, 4 tuning queries, 12 fresh queries (6 present, 6 absent). Each arm/query had three timing repetitions with rotating arm order. Outputs were deterministic across repeats; accuracy counts each case once. Discovery used 24 searches; retest used 234; setup added 72 POSTs. All 258 searches returned HTTP 200. The 330 audit entries reconcile with those requests; no stored private detail appeared in search responses.

The recorded environment was macOS arm64, Python 3.13.7, FastAPI 0.115.12, httpx 0.28.1 and real SQLite. Latency includes in-process ASGI, auth and audit writes; excludes seeding, registration, network, LLM and human time. Model requests = 0, model tokens = 0, model API charges = $0. CPU/electricity cost is not priced. Ten regression tests passed, including two motivated by tuning failures.

## Repeat the comparison and inspect the evidence

The evidence bundle contains the exact source, original matcher snapshot, case labels, pre-run plan, change record, dependency lock, raw JSON responses, audit logs, manifests and an independent verifier. The verifier re-scores saved responses against fixture answers rather than trusting an agent's success claim. Timing will vary across machines.

Unzip the bundle and open `Lost-Link-HW7/HW7_README.md`. Core commands after creating a venv and installing `eval/hw7_requirements.lock`:

```
python -m eval.verify_hw7
python -m pytest -q
python -m eval.hw7_benchmark --phase discovery
python -m eval.hw7_benchmark --phase retest
```

For chronological reruns, follow the README's copy/restore commands to run discovery with the original source first; running the last two commands overwrites local evidence. The submitted evidence preserves the initial pre-change discovery.

**Execution evidence:** readable responses, metrics and audit / source and reproducibility ZIP / repeat instructions. Public GitHub links are accessible without signing in. The same ZIP is embedded in this PDF; open the Attachments panel in a compatible reader, or use the extraction code in the README.

Bundle SHA-256: 4a6a8db99fe0ca3e92b4c79928ee71a43162a1620b57e3617c43e4ab61fbfb34

## What remains unproven

All cases are synthetic and AI-assisted. The original ten queries were already visible and are regression evidence only. Fresh queries were saved before discovery and executed only after the rule was fixed, but the evaluator authored both cases and change: this is not independent human validation. The 50% fresh absence rate is a test design choice, not a measured user distribution. Twelve fresh cases and deterministic repeats give weak population-level evidence.

The conservative rule can miss unfamiliar types and uncertain descriptions; same-type absent items, stale dates, near duplicates, multilingual wording and real campus reports remain under-tested. The study does not establish LLM tool-selection reliability, human verification accuracy, end-to-end item recovery, concurrent capacity, network latency or production performance. Next, test independently collected, blinded cases with unsupported types and same-type hard negatives.